Problem-Solution Fit

Date	21 October 2026
Team ID	TDS-AI-2026
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	4 Marks

Project Title

Personalized Travel Itinerary Planner

Problem-Solution Mapping

#	Problem	Solution	How It's Implemented
1	Travelers spend hours researching and organizing trip plans manually	AI generates a complete multi-day itinerary in 10-30 seconds	Gemini/Groq LLM with structured prompting produces day-by-day plans with activities, times, locations, and costs
2	Generic travel recommendations don't match individual preferences	Personalized itineraries based on interests, travel style, and group size	User inputs (destination, budget, style, interests) are embedded into AI prompts that produce tailored results
3	Difficult to manage and track trip budgets across activities	Automatic cost estimation with per-activity, per-day, and trip-level budget tracking	AI assigns estimated costs to each activity; backend recomputes day and trip totals dynamically
4	Itineraries are rigid and hard to modify once created	Full CRUD on activities + day-level regeneration + drag-and-drop reordering	Edit/delete individual activities, regenerate entire days, reorder via dnd-kit drag handles
5	No secure, private way to store trip plans	User authentication with Row-Level Security ensuring data isolation	Supabase Auth (JWT) + RLS policies ensure each user only accesses their own trips
6	Travel planning tools are overly complex	Clean, intuitive UI with guided form inputs and progressive disclosure	React + shadcn/ui components with clear form labels, validation messages, and loading states
7	AI output may be unreliable or invalid	Robust validation and retry mechanism with automatic fallback	Pydantic schema validation on AI output, retry once on bad JSON, fallback from Gemini to Groq

Value Proposition

For travelers who struggle with the complexity and time investment of trip planning, the Personalized Travel Itinerary Planner is an AI-powered web application **that generates customized, budget-aware, day-by-day itineraries**

in seconds.

Unlike generic travel planning tools, **our product** uses Generative AI to create truly personalized plans with real places, respects the user's budget and travel style, and allows flexible customization through editing, regeneration, and drag-and-drop reordering.

Solution Validation Criteria

Criteria	Status	Evidence
AI generates valid, structured itineraries	Validated	Pydantic GenItinerary model validates all AI output; retry mechanism handles edge cases
Budget is respected in generated plans	Validated	AI prompt explicitly sets total budget and per-day budget limits
Real places are used (not generic)	Validated	AI prompt instructs "Use real places in {destination}"
Users can customize after generation	Validated	Edit, delete, reorder, regenerate features all implemented
System is secure and private	Validated	JWT auth + Supabase RLS + per-user DB client scoping
Performance meets expectations	Validated	AI generation: 10-30 sec; CRUD operations: under 500ms
Application is deployed and accessible	Validated	Frontend on Vercel, Backend on Render, DB on Supabase